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Research Article

Liquid chromatography at critical conditions in ternary mobile phases: Gradient elution along the critical line

In ternary mobile phases consisting of acetone, methanol, and water, the retention of PEG on reversed-phase columns is independent on molar mass at certain compositions of the mobile phase. Along this critical adsorption line, the retention of polypropylene glycol varies quite strongly, which can be utilized in the separation of block copolymers. Gradient elution along the critical line allows a baseline separation of all oligomers in polypropylene glycol up to approximately 25 propylene oxide units. The same resolution can be achieved in the separation of ethylene oxide-propylene oxide block copolymers, regardless of the length of the ethylene oxide block.

Keywords: Block copolymer / Critical conditions / Gradient elution / LC / PEG / Polypropylene glycol / Ternary mobile phase
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1 Introduction

In the analysis of block copolymers, different kinds of information are required for the characterization of a sample: the overall molar mass and its distribution (MMD), the chemical composition and its distribution, the type of functionality and its distribution, moreover the length of the individual blocks and the architecture (di- or triblock, symmetric or asymmetric?). Obviously, this cannot be achieved by simple methods: typically, 2-D methods are required, such as 2-D-LC [1–6] and chromatography coupled with MS, especially with MALDI-TOF-MS [7]. In some cases, combination of chromatography and NMR spectroscopy may be very useful [8–12].

An especially important example for such materials are block copolymers of ethylene oxide (EO) and propylene oxide (PO), which are used for many different applications [13].

Such block copolymers may consist of two or three (or more) blocks. In the triblocks, the hydrophilic block (EO) may form the center of the chain or the arms. In the following text, the abbreviations EO-PO for diblocks and EO-PO-EO or PO-EO-PO for triblocks will be used.

Diblocks can be synthesized by propoxylation of monoethers of PEG, while triblocks are obtained by ethoxylation of polypropylene glycol or by propoxylation of PEG.

One has to keep in mind that triblocks may contain the homopolymers as well as the diblock copolymer as side products. As the physical properties of the block copolymers (and thus their practical performance) are strongly influenced by the distributions mentioned above and the content of the side products [14], there is a strong need for reliable analytical methods, which provide the required information. This would allow optimization of the synthesis in order to yield products of high purity.

Basically, one may apply different modes of LC, which can be characterized by the so-called interaction parameter c [15], which depends on mobile phase composition (for a given polymer-sorbent system), and on temperature.

The most popular chromatographic technique in polymer analysis is size exclusion chromatography (SEC). In SEC, the interaction parameter is strongly negative. SEC yields the MMD [16, 17], and with dual detection [17–23] one may also obtain information on the chemical composition along the MMD. It must, however, be mentioned, that SEC cannot discriminate block copolymers and a mixture of homopolymers.

Liquid adsorption chromatography (LAC) is commonly used in the analysis of low molecular compounds: LAC

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Abbreviations: CAL, critical adsorption line; **CAP,** critical adsorption point; **EO,** ethylene oxide; **LAC,** liquid adsorption chromatography; **LCCC,** LC at critical conditions; **MMD,** molar mass distribution; **MME,** monomethyl ether; **PO,** propylene oxide; **PPG,** polypropylene glycol; **SEC,** size exclusion chromatography

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separates according to the number of groups, which may interact with the surface of the stationary phase. In the analysis of polymers, the selectivity of LAC may be by far too high, as retention increases exponentially with the number of repeat units. Consequently, separation of polymers typically requires gradient elution. In LAC, the interaction parameter is always positive. In the case of copolymers, the situation becomes even more complicated, as two (or more) interaction parameters for the different repeat units have to be taken into account. For A-B polymers, there will be two interaction parameters (c_A and c_B): if both are positive, but with a different value, this will typically result in highly complex peak patterns.

A very useful tool in the characterization of copolymers is LC at critical conditions (LCCC) [24–34]. At the so-called critical adsorption point (CAP) – a special temperature and a special mobile phase composition, where the interaction parameter equals zero – retention is independent of molar mass. Under such conditions, a block copolymer may be separated from a homopolymer.

At the CAP for the block A of an A-B block copolymer (where $c_A = 0$), the block A becomes ‘‘chromatographically invisible’’, and thus the homopolymer elutes at the void volume. The elution volume of the copolymer will then depend on the chemical nature and molar mass of the block B, which may elute in the SEC regime (with $c_B < 0$, *i.e.* earlier than the critical homopolymer) or in the LAC regime (with $c_B > 0$ and elution volumes much larger than the void volume, *i.e.* much later then the critical homopolymer) [27, 34–36]. In the following text, we will use the abbreviations LCCC-E (for $c_A = 0$, $c_B < 0$) and LCCC-A (for $c_A = 0$, $c_B > 0$).

The situation is rather simple in LCCC-E: if the interaction parameter of the excluded block is sufficiently negative, its exact value is not important. In LCCC-A, however, the retention of the adsorbing block (B) may be by far too high at the CAP for the block A: in this case, only the lowest oligomers may elute at all, while the higher ones will be retained on the column.

Such a situation exists in the analysis of EO-PO block copolymers, as we have described in previous papers [27, 34]: A CAP for PEG can be found on typical reversed phases in different mobile phases with different composition [37]. In acetone–water, the interaction parameter of PPG is too large; in methanol–water it is rather too small [27].

LCCC is typically used in isocratic mode, as with gradient elution the critical conditions will be lost. Gradient elution has been proposed for the separation of different homopolymers, which should have ‘‘critical’’ behavior [38, 39], but this approach cannot be applied in LCCC-A.

In some mobile phases, the critical region is very broad, which allows gradient elution in a (limited) range, where the interaction parameter is close enough to zero [40, 41], in others the CAP is very sharp: in such solvents, the composition must be adjusted very precisely.

In a previous paper [42], we have shown that critical conditions exist in ternary mobile phases along a ‘‘critical adsorption line’’ (CAL), which connects the two critical points in the binary mobile phases. In this study, this approach should be applied to gradient separations along the critical line.

2 Materials and methods

EO-PO diblock copolymers were synthesized by microwave assisted anionic ring opening polymerization of PO by using PEG-monomethyl ether (MME) as initiator as has been reported elsewhere [20].

The solvents for HPLC (acetone, methanol, and water, all HPLC grade) were purchased from Roth (Karlsruhe, Germany).

All separations were performed on a gradient system Ultimate 3000, consisting of a pump DGP-3600A, solvent degasser SRD-3600, column thermostat TCC-3000, autosampler WPS-3000SL, all from Dionex (Germerink, Germany), an evaporative light scattering detector PL-ELS 2100 (Polymer Laboratories, Church Stretton, Shropshire, UK). Data acquisition and processing were performed using the software Chromeleon (Dionex, Germerink, Germany). Gradient separations were performed using the following columns:

Synergi MAX RP (silica-based octadecyl phase; 250 × 4.6 mm; particle diameter = 4 μm; nominal pore size = 80 Å, carbon load: 17%, from Phenomenex, Torrance, CA, USA);

Jupiter 5 μm C18 300 Å (silica-based octadecyl phase; 250 × 4.6 mm; particle diameter = 5 μm; nominal pore size = 300 Å, carbon load: 13.34%, from Phenomenex, Torrance, CA, USA).

Mobile phases were mixed by the gradient pump. All isocratic chromatograms were obtained at a flow rate of 0.5 mL/min, the gradient chromatograms on the Jupiter column at a flow rate of 1.0 mL/min.

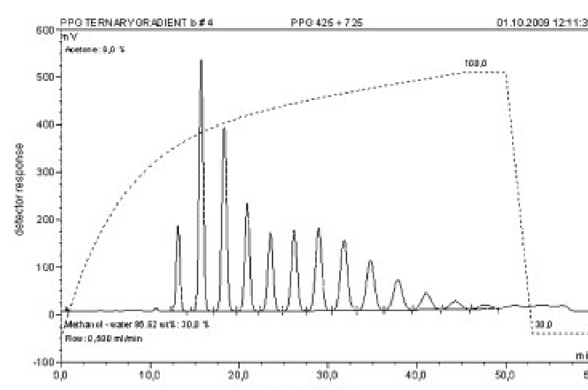

Figure 1. Gradient elution along the CAL. Synergi MAX RP, 0.5 mL/min, solvent A: 26.90 wt% acetone, solvent B: 86.82 wt% methanol, start: 70% A, then in 45 min to 100% B, 5 min constant at 100% B (see Table 1).

3 Results and discussion

As we have shown in previous papers [40, 41], critical conditions can be found in some mobile phases not at a sharp point, but rather as a plateau. For example, the interaction parameter of PEG on typical reversed-phase columns in methanol–water mobile phases is quite close to zero in a range between 75 and 95 wt% methanol [41], and a similar wide range is also observed in acetonitrile–water (beween 35 and 60 wt% acetonitrile) [43]. Consequently, one may even run a gradient in order to improve the LAC separation according to the PO block and still be close to critical conditions for the EO block. This may work quite well, as long as the EO block is not too large [41].

The problem can, of course, be solved much better by using ternary mobile phases, in which the interaction parameter of the adsorbing block can be adjusted while having really critical conditions for the other one: In a previous paper [42],] we have determined the interaction parameter of PEG in the LAC range on reversed-phase columns in ternary mobile phases and extrapolated to c(PEG) = 0. It was found that critical conditions for PEG existed along a straight line between the critical points in binary mobile phases. Isocratic separations showed good agreement of the chromatograms obtained with PPG and EO-PO diblock copolymers.

In this study, we have determined the CAP in binary and ternary mobile phases precisely: on the Synergi MAX RP column critical conditions for PEG were observed in methanol–water containing 86.8 wt% methanol and in acetone–water containing 26.9 wt% acetone.

In the first step, it was assumed that critical conditions for PEG should exist along a straight line connecting these compositions. Hence, we tried to separate PPGs by gradient elution along this line under the conditions described previously [42].

As can be seen in Fig. 1, a baseline separation of all oligomers in a mixture of PPG 425 and 725 could be achieved using the gradient described in Table 1.

When we analyzed two diblock copolymers initiated with PEG-MME 350 and 1100 and a monomer-to-initiator ratio of 1:1, well-resolved chromatograms were obtained, as can be seen in Fig. 2.

The molar mass range, which could be analyzed under these conditions, was, however, not wide enough. Hence, we tried to apply another gradient starting at another composition on the CAL (Table 2), and obtained a

Table 1. Gradient along the CAL for PEG on Synergi MAX RP

	Acetone–water 26.90 wt%	Methanol–water 86.82 wt%	
Time (min)	A	B	Curve shape
0	70%	30%	2
45	0%	100%	Linear
50	0%	100%	Linear
53	70%	30%	Linear
60	70%	30%	Linear

Table 2. Gradient along the CAL for PEG on Synergi MAX RP

	Acetone–water 26.90 wt%	Methanol–water 86.82 wt%	
Time (min)	A	B	Curve shape
0	50%	50%	
45	0%	100%	2
50	0%	100%	Linear
53	50%	50%	Linear
60	50%	50%	Linear

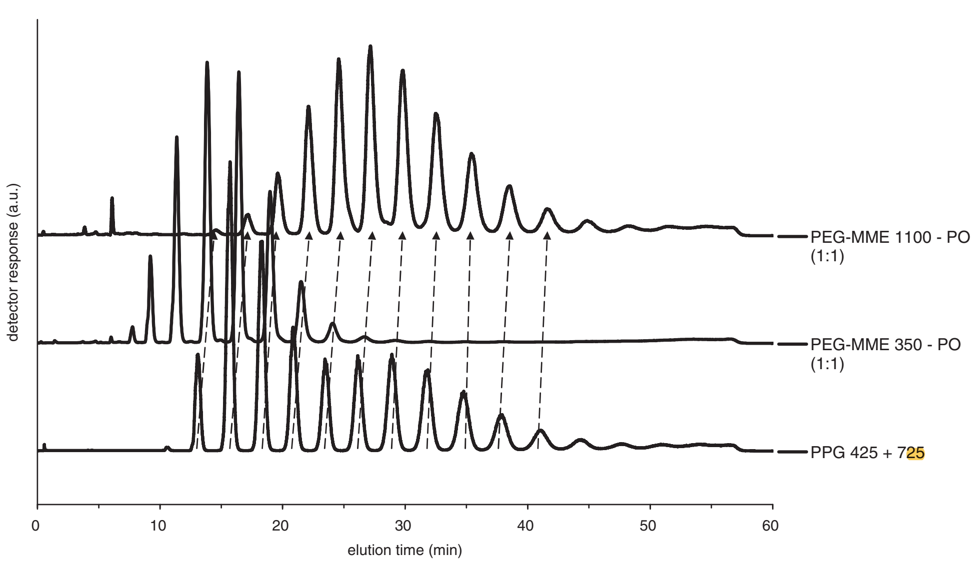

Figure 2. Gradient elution of a PPG mixture (PPG 425 and 725) and two EO-PO diblock copolymers (conditions as above).

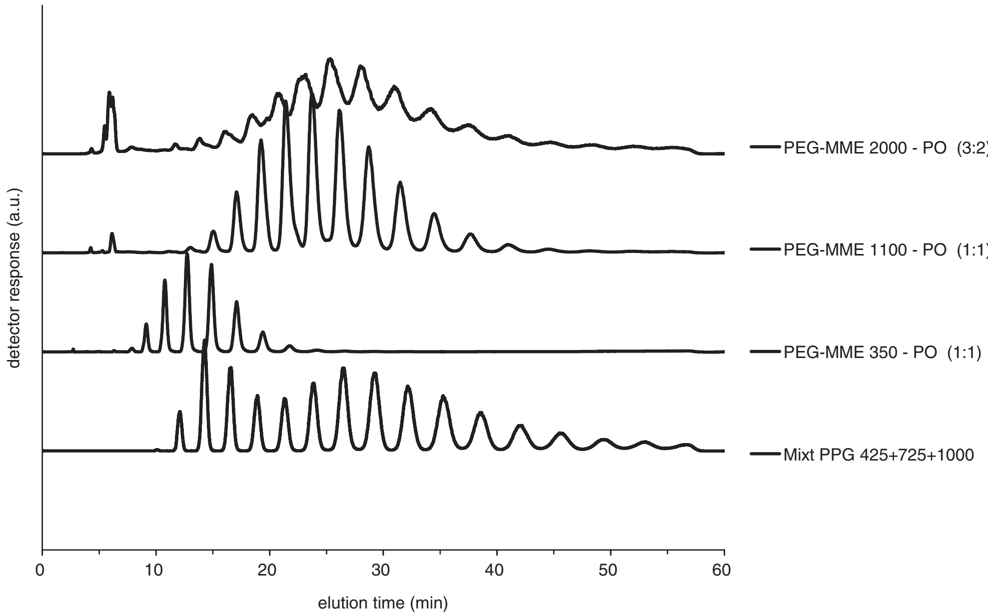

Figure 3. Gradient elution of a PPG mixture (PPG 425 and 725) and three EO-PO diblock copolymers (conditions see Table 2).

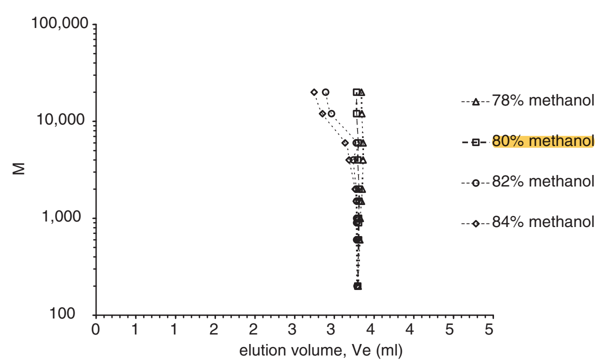

Figure 4. Searching critical conditions for PEG on the Jupiter C18 300 Å column in binary mobile phases: methanol–water.

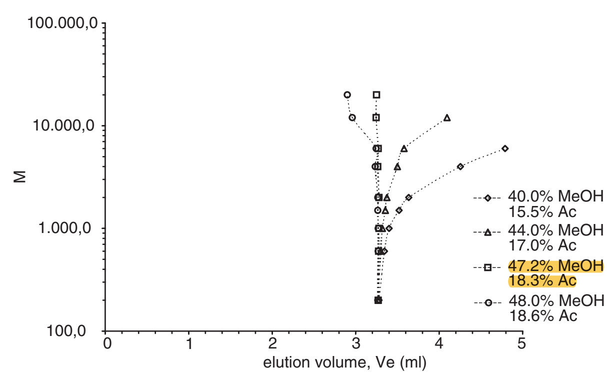

Figure 6. Searching critical conditions for PEG on the Jupiter C18 300 Å column in ternary mobile phases: acetone–methanol–water.

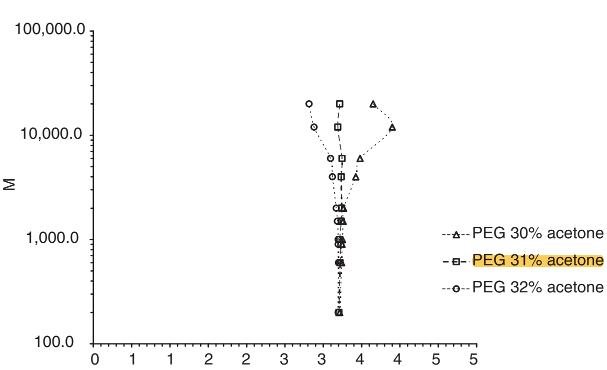

Figure 5. Searching critical conditions for PEG on the Jupiter C18 300 Å column in binary mobile phases: acetone–water.

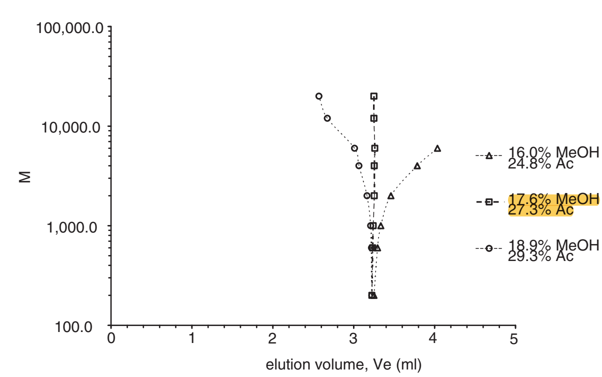

Figure 7. Searching critical conditions for PEG on the Jupiter C18 300 Å column in ternary mobile phases: acetone–methanol–water.

reasonable separation even of PPG 1000, but a disappointing result for a diblock copolymer initiated with PEG-MME 2000 and a monomer-to-initiator ratio of 3:2 (Fig. 3). This may have two different reasons: first of all, the pore width of the Synergi MAX RP column (80 Å) is too small. The other explanation may be that the gradient along the straight line

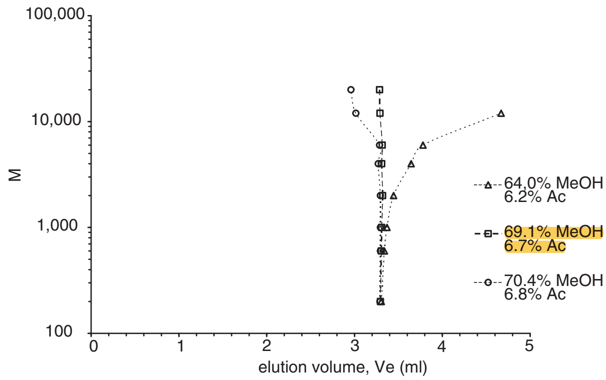

Figure 8. Searching critical conditions for PEG on the Jupiter C18 300 Å column in ternary mobile phases: acetone–methanol –water.

connecting the critical points in the binary mobile phases may not follow the real CAL.

Consequently, we have exchanged the narrow pore column by another one with wider pores (Jupiter 5 μm C18 300 Å) and determined the critical compositions in the binary and in several ternary mobile phases.

As can be seen in the following figures, exactly critical behavior was observed up to the highest PEG used in this study (PEG 20000) in binary mobile phase containing 80 vol% (Fig. 4) or 31 vol% acetone (Fig. 5).

In the ternary mobile phases, however, critical conditions were found in mobile phase conditions containing 44.0 vol% methanol and 17.0 vol% acetone (Fig. 6) instead of the expected 40% methanol and 15.5% acetone, which should be critical if the CAL would be a straight line. Moving left and right, we found critical conditions at

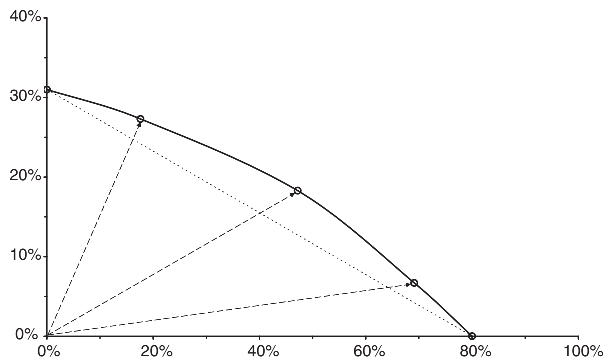

Figure 9. Critical adsorption line for PEG on the Jupiter C18 column.

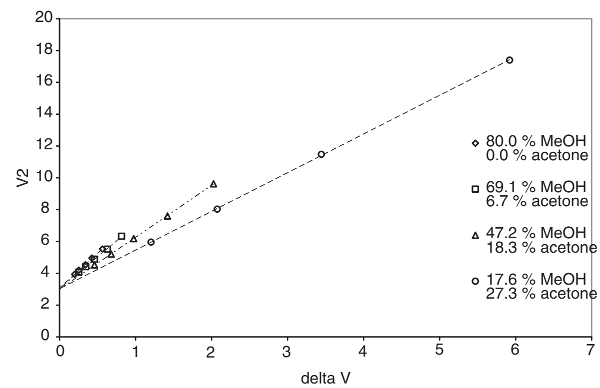

Figure 11. Determination of accessible volume and interaction parameter of PPG on Jupiter C18 in mobile phases compositions along the CAL (see Fig. 10).

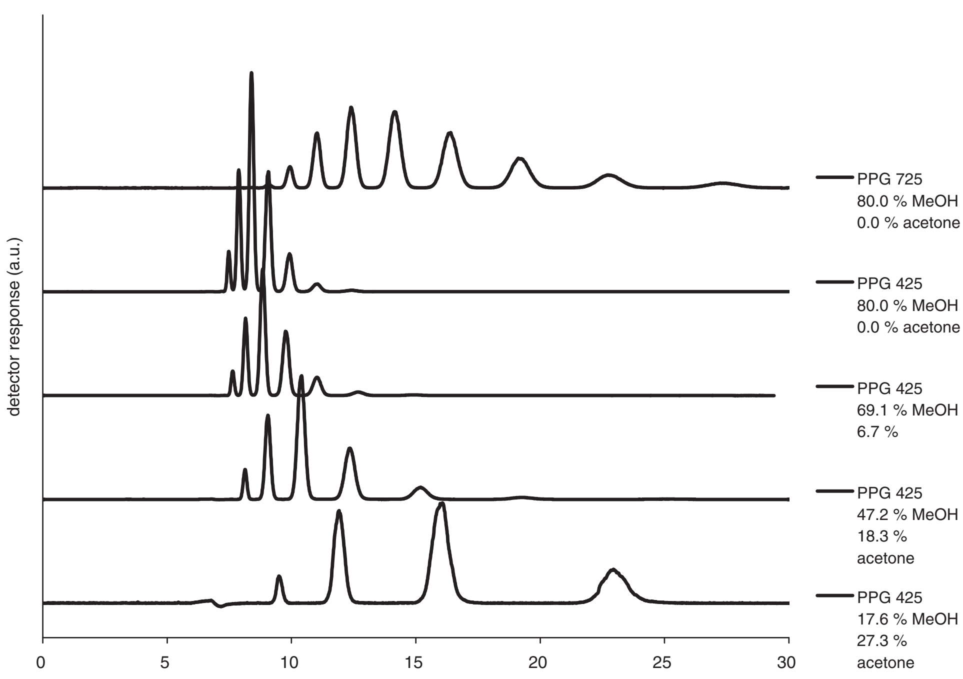

Figure 10. Isocratic elution of PPG 425 on Jupiter C18 in mobile phase compositions along the CAL.

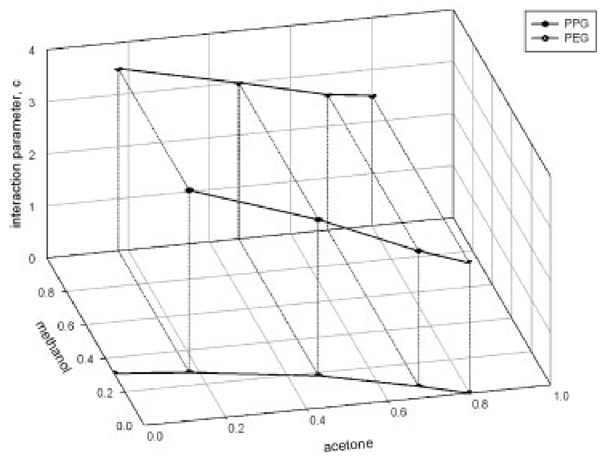

Figure 12. Interaction parameter of PPG on Jupiter C18 in mobile phase compositions along the CAL (see Figs. 10 and 11).

Table 3. Gradient along the CAL for PEG on Jupiter C18 300 Å

Min	Methanol	Acetone	Curve shape
0	17.60%	27.30%	
3	47.20%	18.29%	Linear
6	69.12%	6.70%	Linear
10	80.00%	0.00%	Linear
15	85.00%	0.00%	Linear
30	90.00%	0.00%	Linear
35	90.00%	0.00%	Linear
40	17.60%	27.30%	Linear
45	17.60%	27.30%	Linear

Flow rate: 1.0 mL/min.

17.6 vol% methanol and 27.3 vol% acetone (Fig. 7) or 69.1 vol% methanol and 6.7 vol% acetone (Fig. 8).

Evidently, all critical ternary phases contained less water than one would expect if the CAL would be a straight line between the binary CAPs (Fig. 9).

We have now separated PPG 425 in ternary mobile phases corresponding to critical conditions for EO and found that retention could thus be varied in a wide range. From the chromatograms shown in Fig. 10, we have determined the interaction parameters according to a procedure described previously [37].

As can be seen in Fig. 11, straight lines are obtained in a plot of the elution volumes *versus* the difference in elution volumes of consecutive oligomers. From the slope in a such a plot one may calculate the interaction parameter [37]. It must be mentioned that this does not require any information on the number of repeat units for each peak.

Figure 12 shows a three-dimensional representation of the interaction parameter of PPG along the CAL for PEG. We have now used a gradient along this line for the separation of PPGs and EO-PO block copolymers. The profile is given in Table 3.

Obviously, in a binary mobile phase containing 80 vol% methanol (which corresponds exactly to the CAP), higher oligomers of PPG would elute very late. As mobile phases with a higher methanol content are still almost critical, we followed the CAL up to 80 vol% methanol, and thereafter we increased the methanol content up to 90 vol%.

As can be seen in Fig. 13, a baseline resolution of all oligomers (with respect to the number of PO units) is achieved under these conditions. From the specification given by the producers of these samples one could expect,

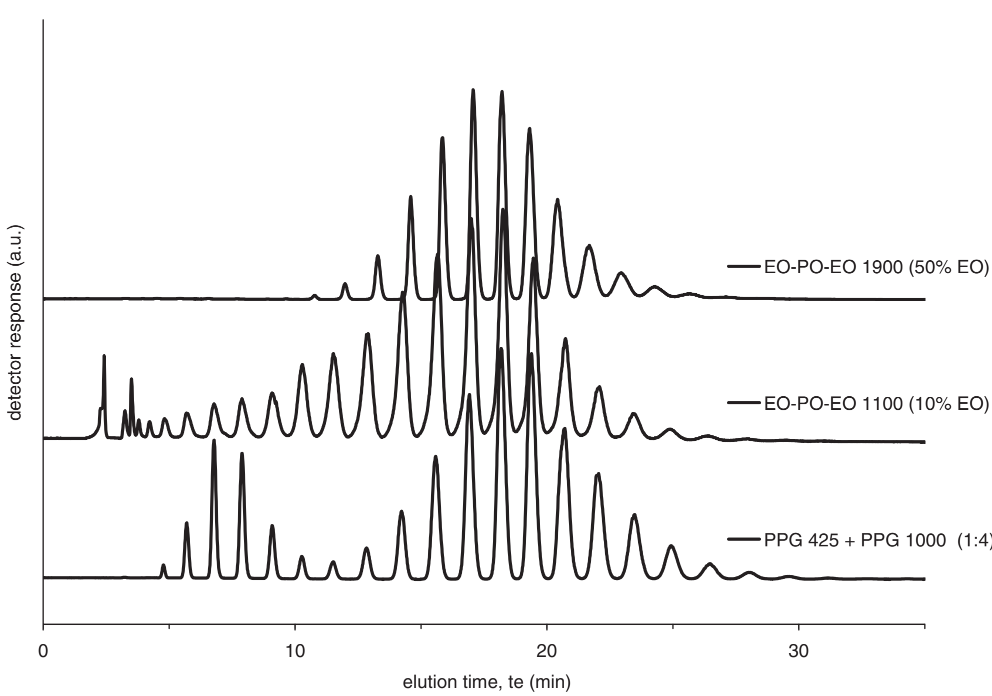

Figure 13. Gradient LCCC of a PPG mixture (PPG 425 and 1000) and two triblock copolymers on the Jupiter C18 column. For gradient conditions, see Table 3.

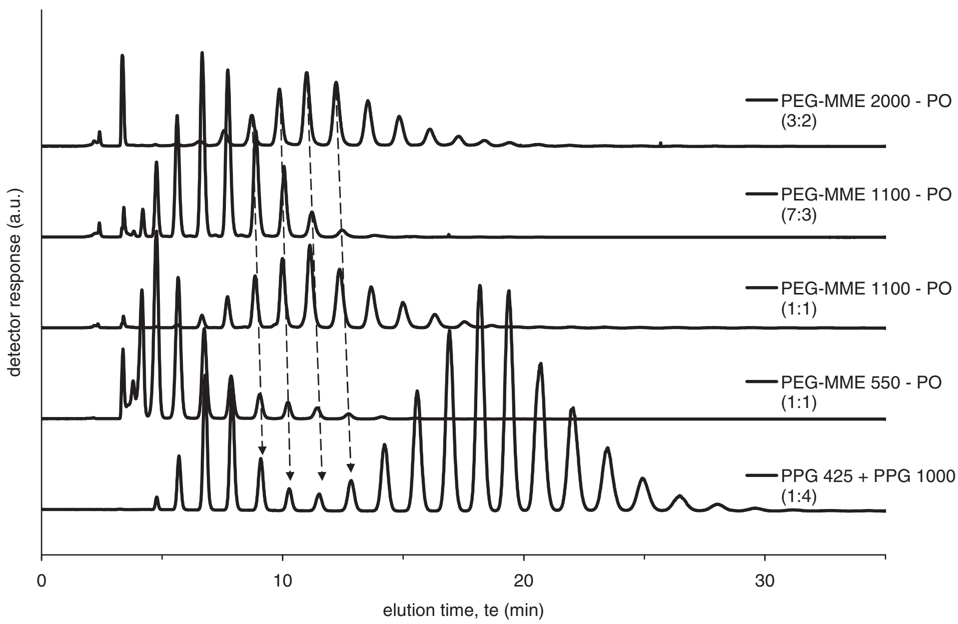

Figure 14. Gradient LCCC of a PPG mixture (PPG 425 and 1000) and three EO-PO diblock copolymers on the Jupiter C18 column; for conditions, see Table 3.

that the MMD of the PO block in both block copolymers should be similar to that of PPG 1000, but this is evidently not the case.

In EO-PO-EO 1100, the maximum of the MMD appears at slightly lower molar mass; moreover, this sample contains a lot of lower oligomers. This allows some conclusions concerning the synthesis and the properties of the product. Evidently, the PO block in EO-PO-EO 1100 is not PPG 1000.

Under the same conditions, we have also analyzed diblock copolymers prepared using PEG-MMEs with different MMD as initiators. As can be seen in Fig. 14, a baseline separation of the individual oligomers (with respect to the number of PO units) is achieved, and the retention is independent on the size of the hydrophilic block.

4 Concluding remarks

On reversed-phase columns, critical conditions for PEG exist in ternary phases consisting of acetone, methanol, and water along a CAL, which is not a straight line, but shows a curvature toward lower water content. The interaction parameter of PPG varies quite strongly along the critical line for PEG. Consequently, gradient elution along the critical line allows a baseline separation of all oligomers in PPG up to approximately 25 PO units. The same resolution can be achieved in the separation of EO-PO block copolymers. In such separations, the elution volumes do not depend on the length of the EO block.

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